

Meta-in-context Learning In large language models

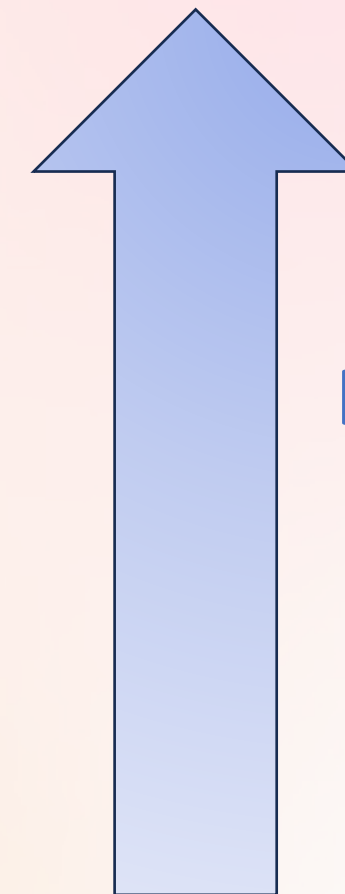
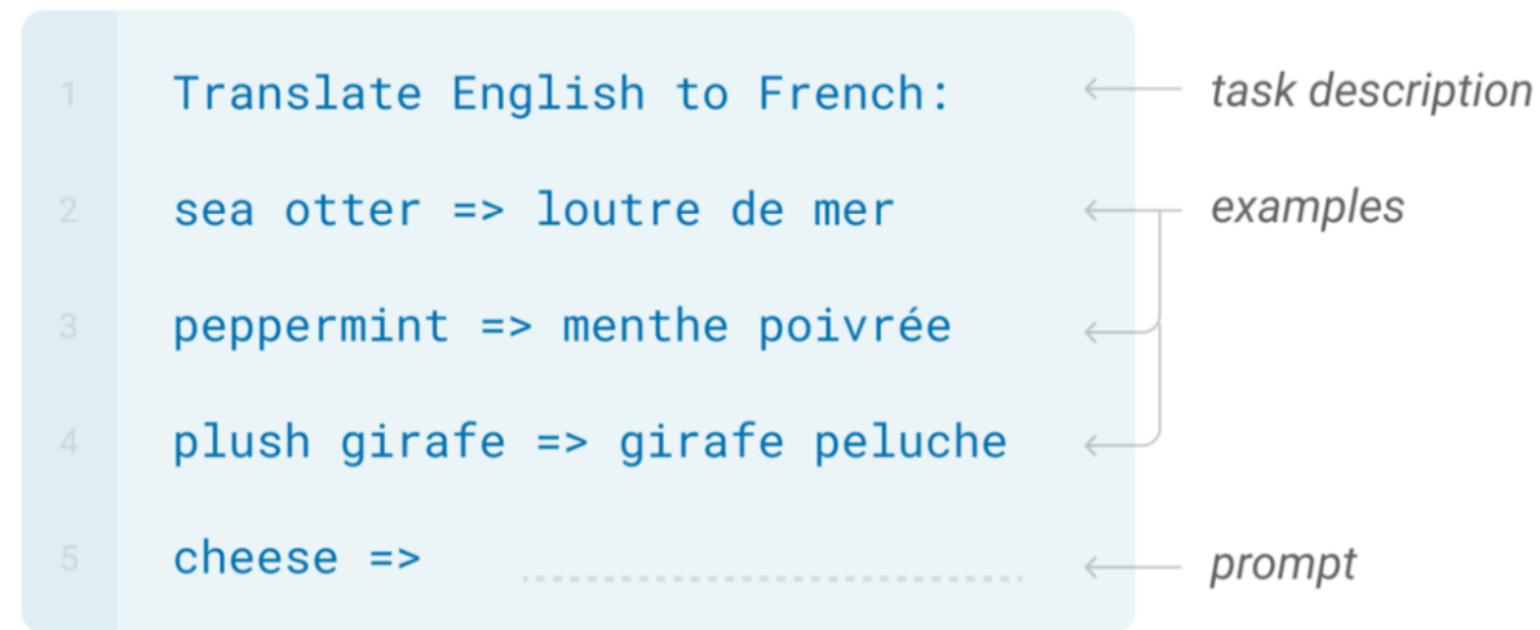
자연어 처리팀: 박주혜(발표자), 김세진, 김영민, 조해창, 박산희

Index

1	Introduction
2	Background
3	Experimental analyses
4	Discussion

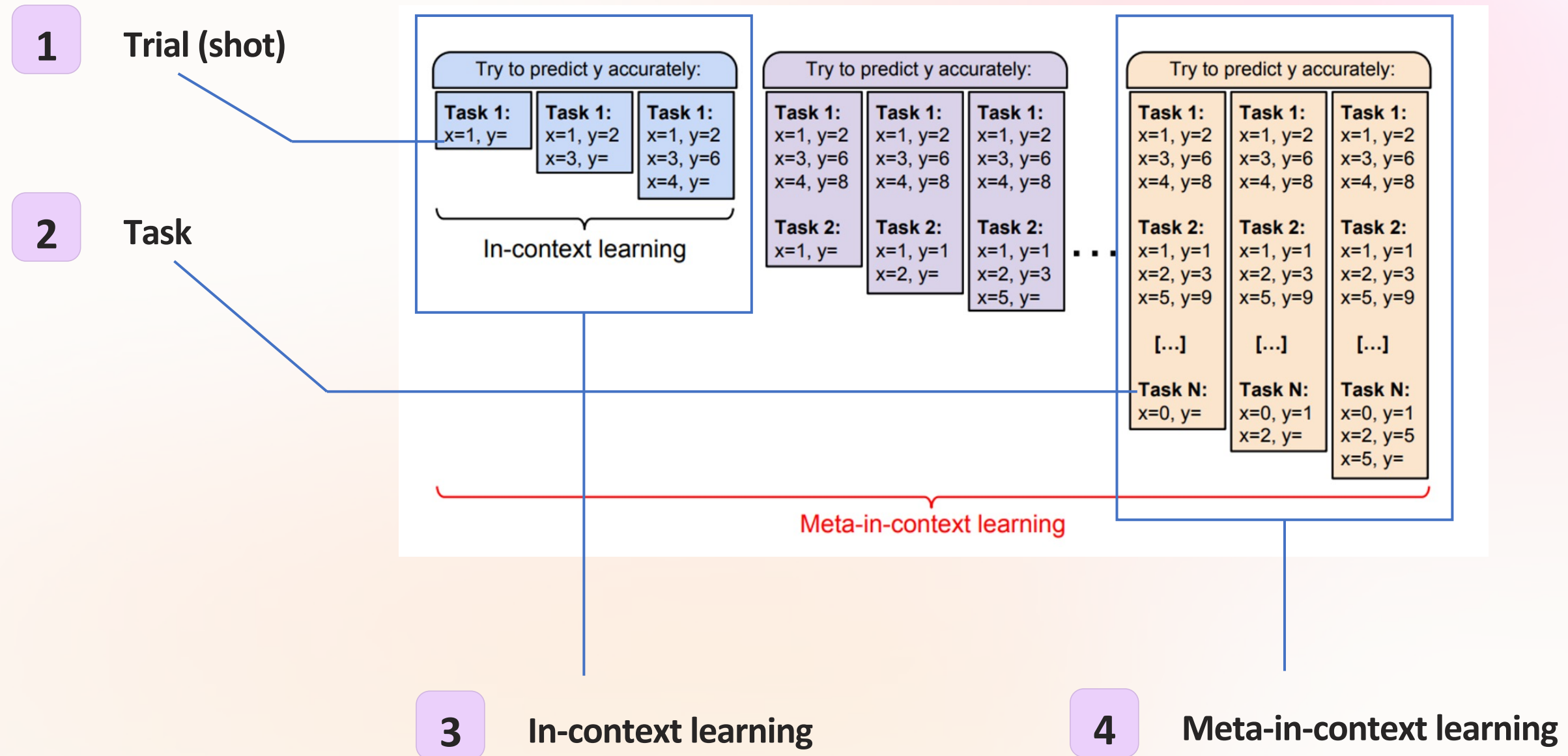
Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



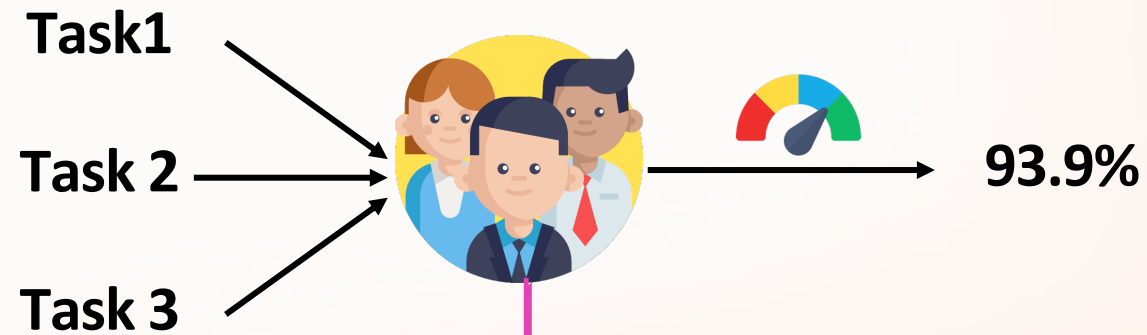
**meta-in-context
learning**

Figure 2.1 from [OpenAI's GPT-3 paper](#)

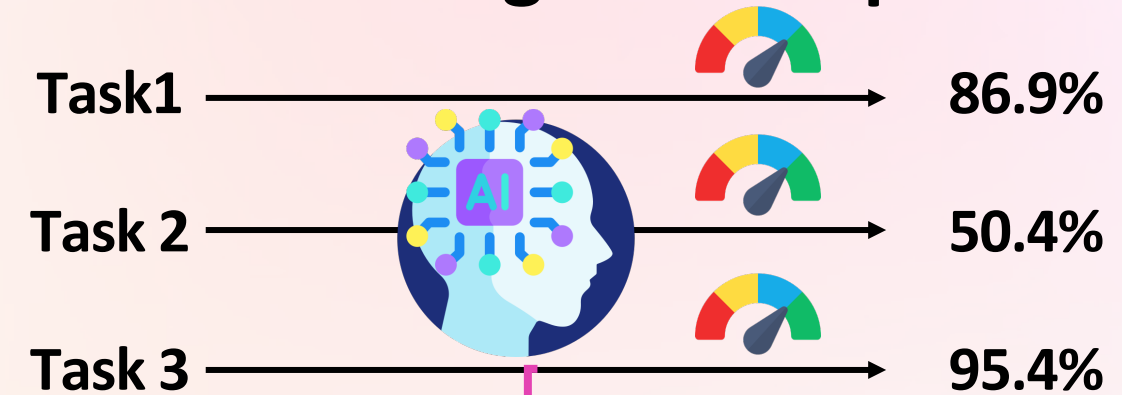


Performance Evaluation Methods

Human – Single Score



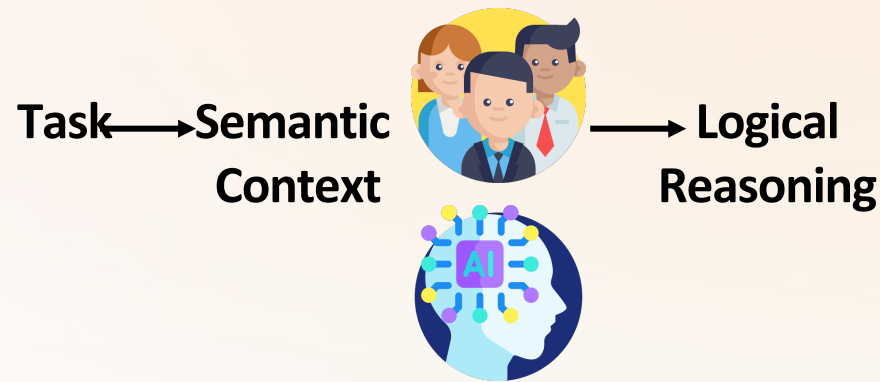
Artificial Intelligence – Separate Scores



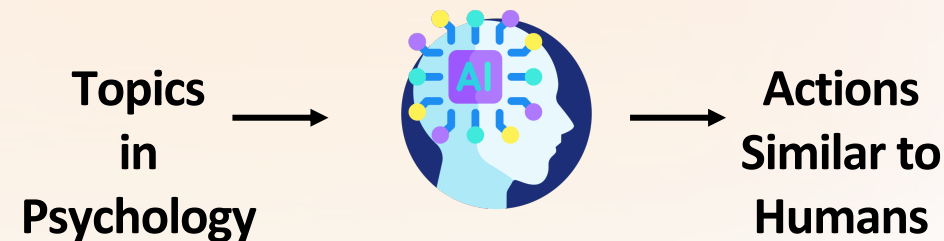
Meta In-Context Learning

Related Works – LLM act similar to humans

Semantic Context in Tasks
(Dasguta et al., 2023)



“Machine Psychology”
(Binz et al., 2023)



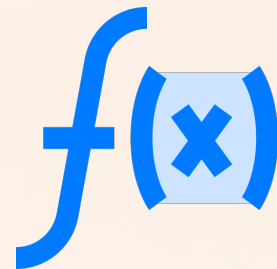
LLM hands recursively nested grammatical structures
(Lampinen et al., 2023)



Model: GPT-3 (TEXT-DAVINCI-002)

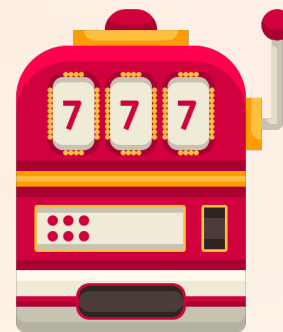
1

one-dimensional functions



2

two-armed bandit tasks



3

Regression on real-world data



One-dimensional functions

Function learning prompt

You observe 5 machines that produce an output y for a given input x .
Each machine implements a different function.

Machine 1:

[...]

Machine 5:

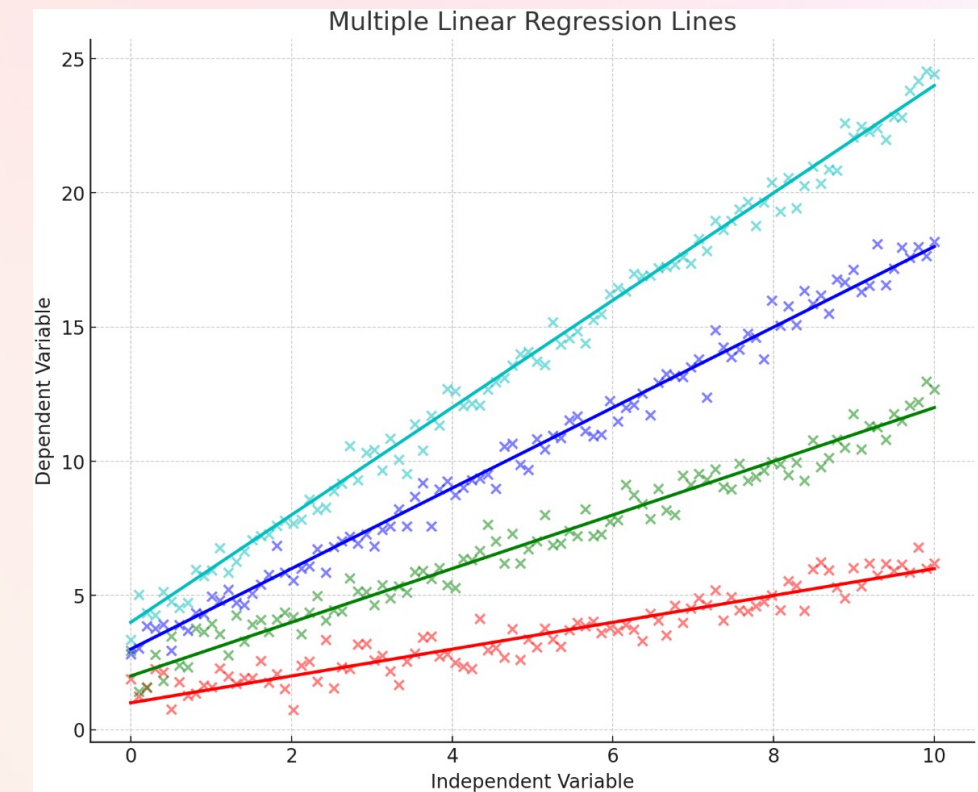
$x = 52, y = -209;$

$x = 18, y = -138;$

$x = 60, y = [\text{insert}];$

Trial (shot)

Task



One-dimensional functions

GPT-3 does in-context learning

- A의 실선 그래프

GPT-3 does meta-in-context learning

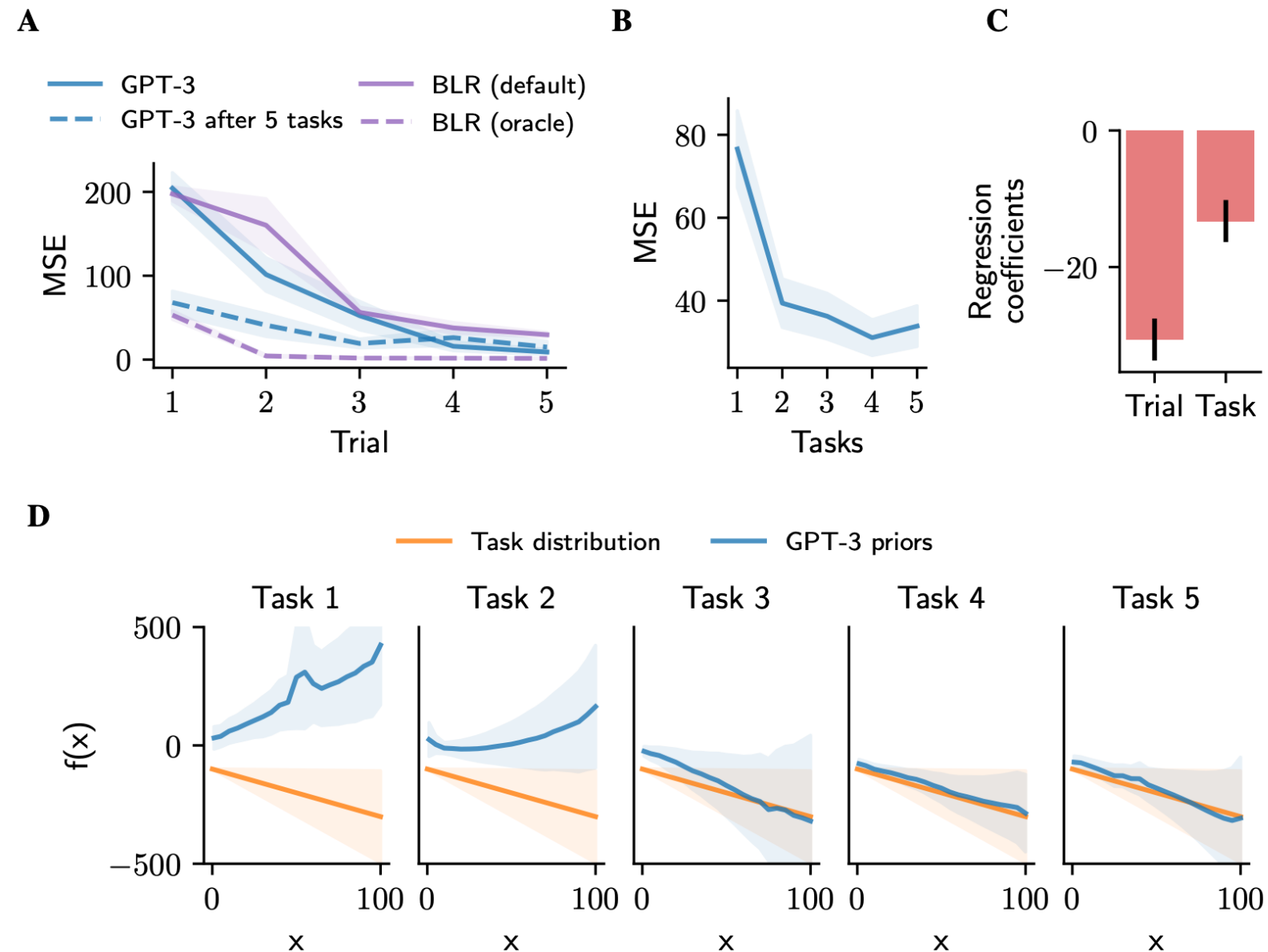
- A의 점선 그래프

- B

- C

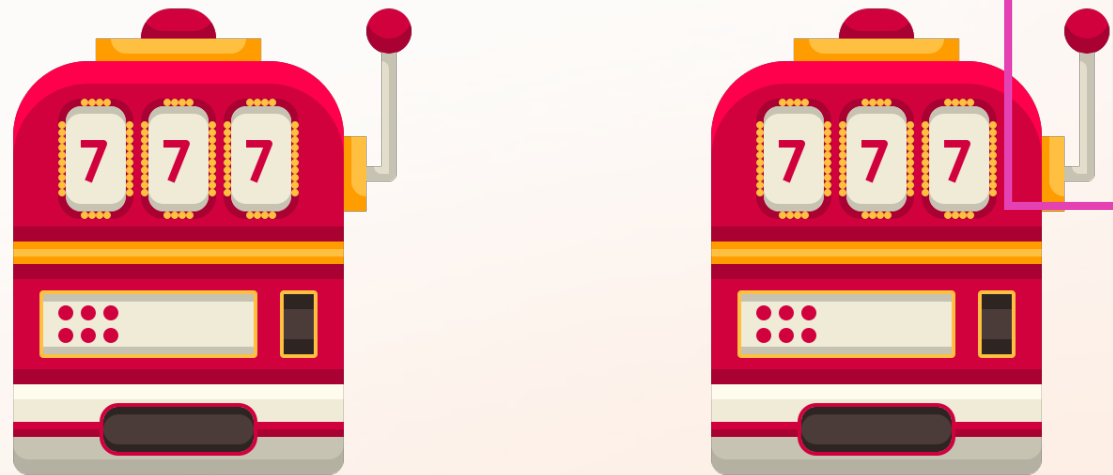
Meta-in-context learning is driven by adaptation of priors

- D



Two-armed bandit tasks

Arm



Bandits

- 여러가지 선택지가 있을 때,
어떤 선택이 가장 큰 보상을 얻을 수 있는지 모를 때,
최적의 전략을 세우기 위한 알고리즘

- 아직 잘 모르는 선택지를 시도해보는 탐험(Exploration)과
높은 보상을 얻을 수 있다고 알려진 선택지를 활용(Exploitation)을
적절히 활용해야 하는 Task

Exploitation

OR

Exploration

“그게 종단 말이지?
묻고 더블로 가!”



아직 안 살펴
본 애들을 좀 더
보면서 가자!



Two-armed bandit tasks

Two-armed bandit task prompt

You are going to different casinos that own two slot machines. Choosing the same slot machine will not always give you the same points, but one slot machine is always better than the other. Within a casino, your goal is to choose the slot machine that will give you the most points over the course of 10 trials. Each casino owns a different pair of machine.

You have received the following points when playing in casino 1:

[...]

You have received the following points when playing in casino 5:

- Machine J delivered 4.2 points.
- Machine F delivered -7.4 points.
- Machine J delivered 3.2 points.
- Machine J delivered 3.9 points.

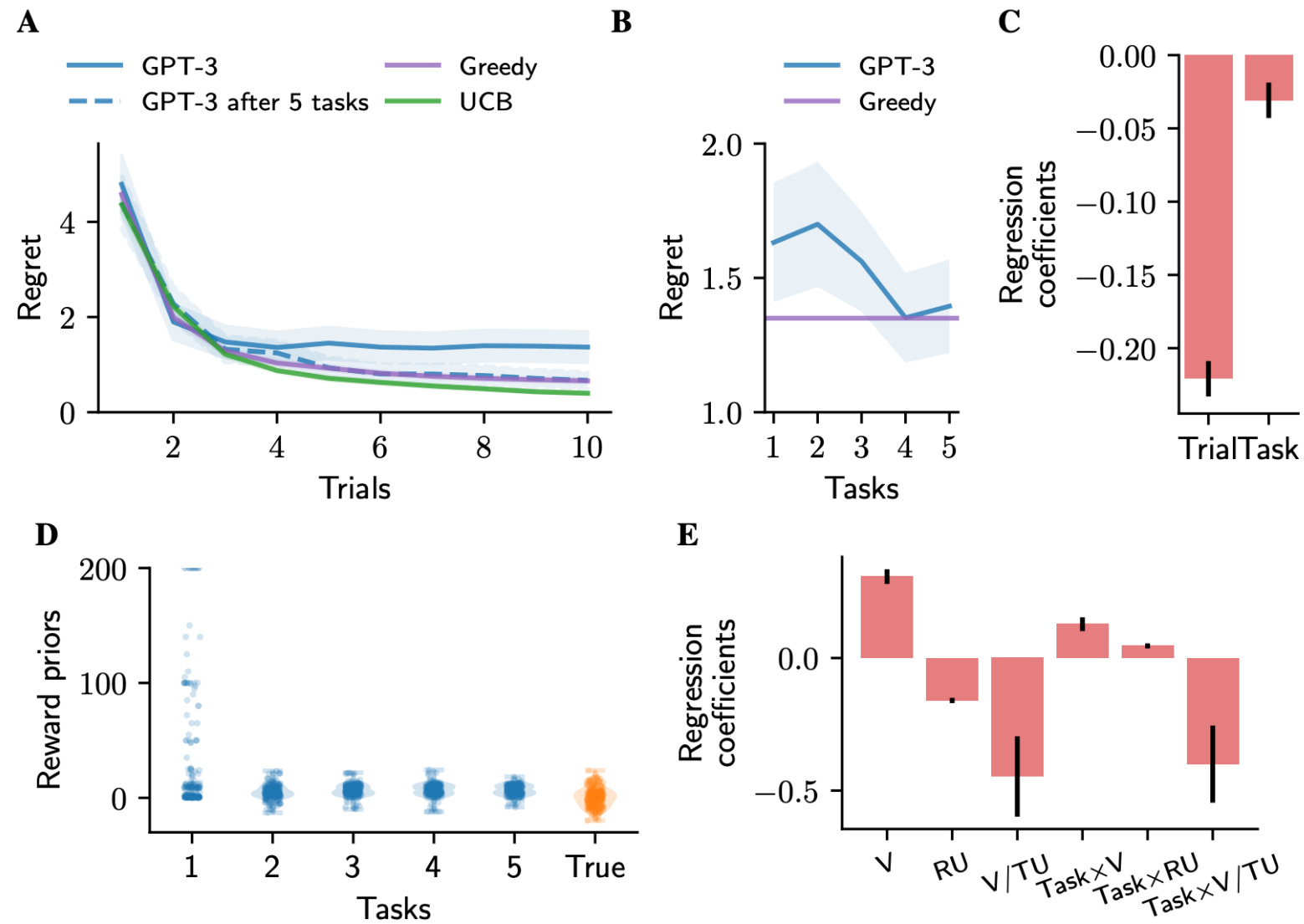
Q: We are now performing trial 5 in casino 5. Which machine do you choose between machine J and machine F?

A: Machine [insert].

Trial (shot)

Task

Two-armed bandit tasks



GPT-3 does in-context learning

- A의 실선 그래프

GPT-3 does meta-in-context learning

- A의 점선 그래프

- B

- C

Meta-in-context learning is driven by adaptation of priors

- D

Meta-in-context learning reshapes learning strategies

- E

Regression on real-world data

ID	Name	Obs.	Predictors	Id	Name	Obs.	Predictors
1	Abalone	4,177	8	31	Land	67	4
2	AFL	41	5	32	Lung	654	4
3	Air	41	6	33	Mammal	58	7
4	Airfoil	1,503	5	34	Medical expenditure	5,574	14
5	Algae	340	11	35	Men	34	3
6	Athlete	202	8	36	Mileage	398	7
7	Basketball	96	4	37	Mine	44	4
8	Birth weight	189	8	38	Monet	430	4
9	Body fat	252	13	39	Mortality	60	15
10	Bone	42	6	40	Movie	62	12
11	Car	93	21	41	Mussel	44	8
12	Cigarette	528	7	42	News	39,644	52
13	Concrete	1,030	8	43	Obesity	136	11
14	Contraception	152	6	44	Occupations	36	3
15	CPU	209	6	45	Pinot	38	6
16	Crime	47	15	46	Pitcher	176	15
17	Diabetes	442	10	47	Plasma	315	12
18	Diamond	308	4	48	Prefecture	45	5
19	Dropout	63	17	49	Prostate	97	8
20	Excavator	33	4	50	Reactor	32	10
21	Fish	413	3	51	Rebellion	32	6
22	Fuel	51	5	52	Recycle	31	7
23	Gambling	47	4	53	Rent	2,053	10
24	Highway	39	11	54	Salary	52	5
25	Hitter	263	19	55	SAT	50	4
26	Home	3,281	4	56	Schooling	3,010	22
27	Homeless	50	7	57	Tip	244	6
28	Infant	101	3	58	Vote	159	5
29	Labor supply	5,320	5	59	Wage	4,360	10
30	Lake	69	10	60	White wine	4,898	11

Table 2: Data sets used in the empirical comparison.

Table 2 from [Simple Regression Models](#)

Regression on real world data

You observe an input vector x and have to predict the corresponding output y as accurately as possible. You are given 5 different tasks.

Task 1:

[...]

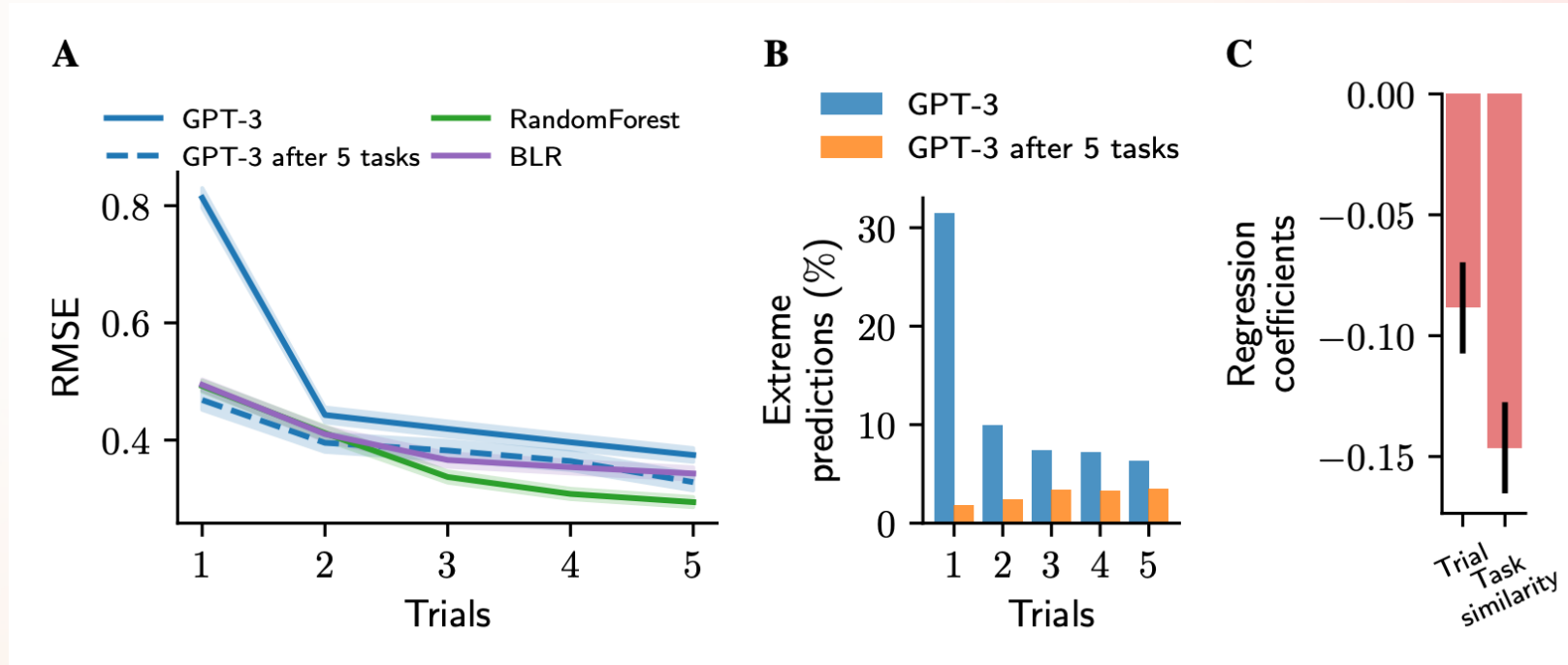
Task 5:

$x = [-0.81, -0.16, -0.78, -0.77, -0.45], y = -0.34;$
 $x = [-0.81, -0.63, -0.75, -0.83, -0.55], y = -0.68;$
 $x = [-0.79, -0.71, -0.76, -0.8, -0.45], y = -0.75;$
 $x = [-0.2, -0.3, -0.28, -0.11, -0.18], y = 1.0;$
 $x = [-0.97, -0.92, -0.97, -0.97, -0.82], y = [\text{insert}];$

Task

Trial (shot)

Regression on real-world data



GPT-3 does in-context and meta-in-context learning

- A

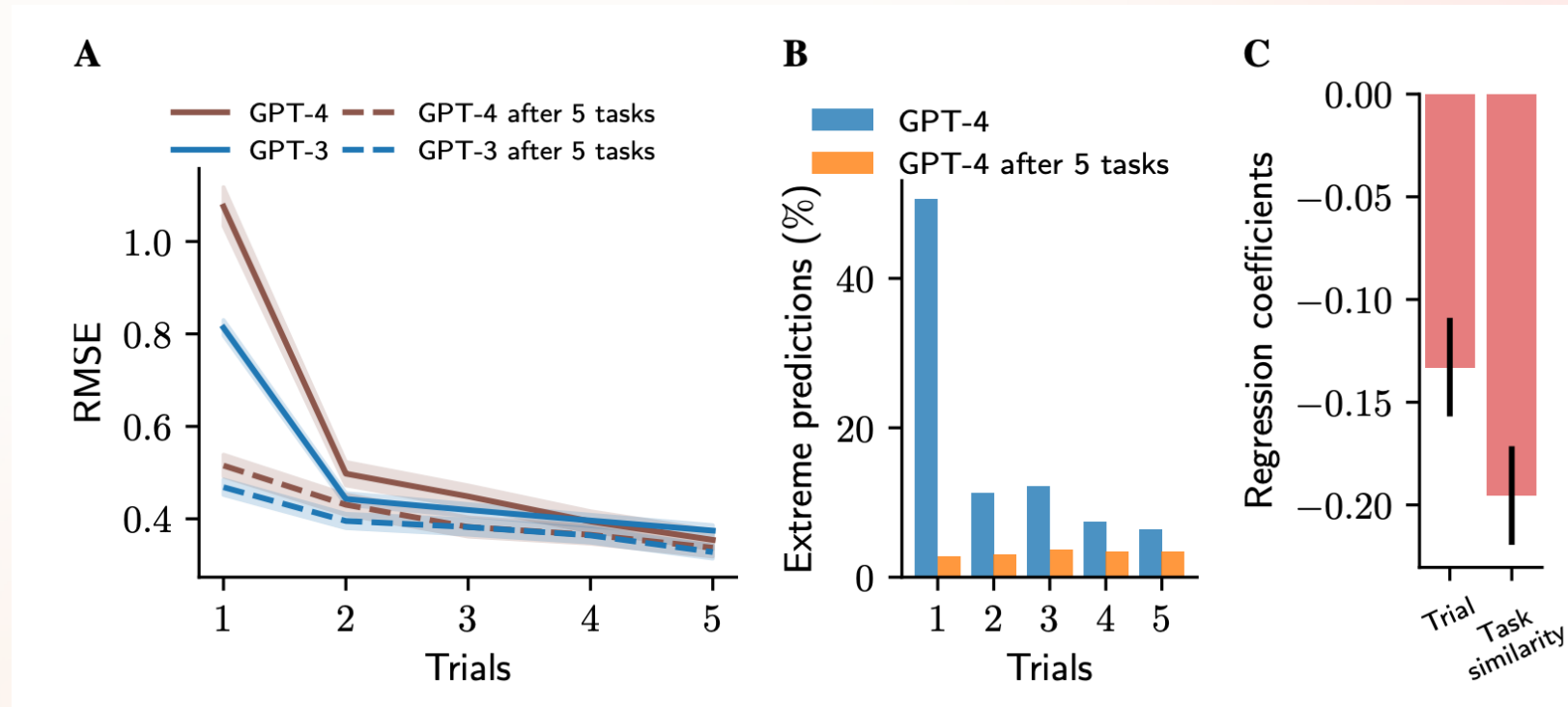
Meta-in-context learning constrains predictions

- B

Meta-in-context learning works better when task similarity is high

- C

Regression on real-world data(GPT-4)



GPT-4 does in-context and meta-in-context learning

- A

Meta-in-context learning constrains predictions

- B

Meta-in-context learning works better when task similarity is high

- C

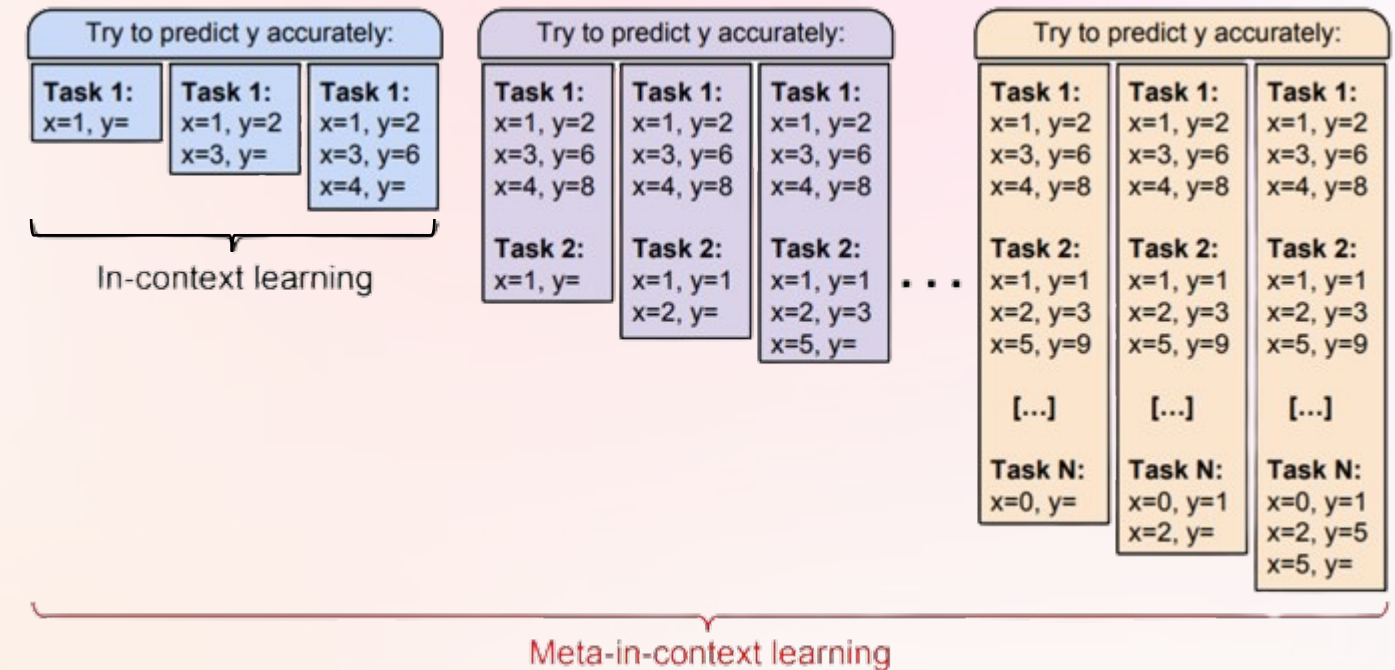
Meta In-context Learning

Meta in-context learning enables LLMs to increase in-context learning capabilities



Through meta in-context learning...

1. The LLM's priors were overwritten
2. Change its learning strategies



Shortcomings of model (Limitations)

1. Used only a handful of conversations
(Limited context window)
2. Tasks were too simplistic in nature
(Did not cover complicated tasks)